College of Computing and Information Sciences

جامعة التقنية والعلوم التطبيقية
University of Technology and Applied Sciences

[COURSE CODE] · [Course title]

# Lecture [N]: [Lecture topic in sentence case]

[Lecturer name] · UTAS-[Branch]

[Semester] [Academic year, e.g. 2026/2027]

Acknowledgment: [Adapted from slides by … / Original work by the course team]. [AI-assistance disclosure, if any.]

Credits

# This lecture builds on the work of others, acknowledged below

| | |
|---|---|
| **Adapted from** | slides by [Name] ([UTAS branch], [Semester]); [what you changed] |
| **Textbook slides** | instructor slides for [Textbook, edition], used under adopter terms |
| **Open materials** | [Course, university], [CC license]; shared under the same license |
| **AI assistance** | [Draft headlines and practice questions prepared with an AI assistant; content reviewed and verified by the lecturer] |

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Learning outcomes

# By the end of this lecture, you should be able to:

1. **Trace** [the execution of … on …] **CLO [x]**
2. **Compare** [… and … in terms of …] **CLO [x]**
3. **Implement** [… to solve …] **CLO [x]**

**Last lecture:** [One line linking to the previous lecture, e.g. the question it left open]

Agenda

# Today’s lecture moves through two parts, one for each outcome

| | | |
|---|---|---|
| **Part 1** | **Searching sorted data** | Outcome 1 |
| **Part 2** | **Choosing a search strategy** | Outcome 2 |
| **Close** | Summary, next steps and references | |

Part 1 of 2

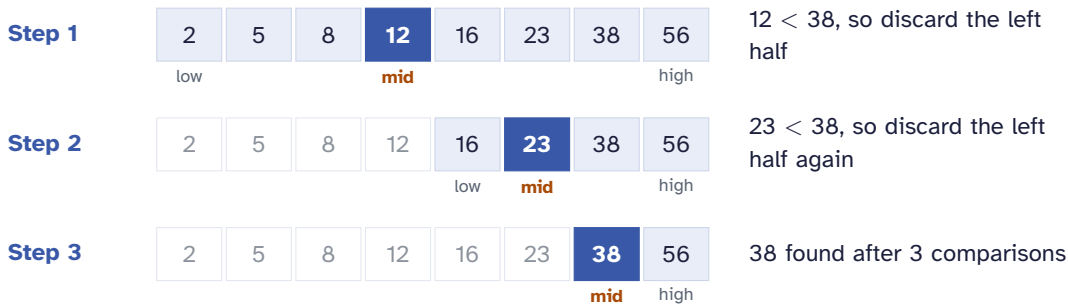
# Searching sorted data

How can we find an item in a sorted array without checking every element?

**Part 1 · now**
**Searching sorted data**

Part 2
Choosing a search strategy

# Binary search halves the search space at each step, so it runs in $O(\log n)$ time




*Note.* Diagram by the course team. Target value: 38; shaded cells are the range still searched.

# Each pass of the loop compares the target with the midpoint and keeps only one half

```
1  def binary_search(items, target):
2      low, high = 0, len(items) - 1
3      while low <= high:
4          mid = (low + high) // 2
5          if items[mid] == target:
6              return mid
7          if items[mid] < target:
8              low = mid + 1
9          else:
10             high = mid - 1
11     return -1
```

**Line 4**

The midpoint splits the remaining range, so each pass discards half of it.

Check your understanding · Part 1

# A sorted array holds 1,024 items. At most how many comparisons does binary search need?

**A** 10

**B** 11

**C** 512

**D** 1,024

Think alone for 1 minute, then compare with a neighbor.

**Next:** Binary search needs sorted data. What does sorting cost?

Part 2 of 2

# Choosing a search strategy

Binary search needs sorted data. What does sorting cost?

# Binary search is faster than a linear scan, but only once the data is sorted

## Linear scan

$O(n)$ comparisons

Works on unsorted data

No preparation needed

## Binary search

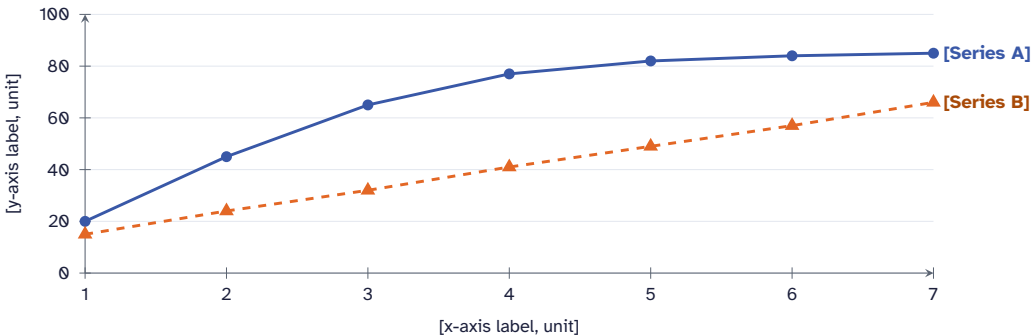
$O(\log n)$ comparisons

Needs sorted data

Sorting first costs $O(n \log n)$

*Note.* [Source line for any adapted material.]

# [Assertion the chart proves, e.g. “Series A levels off while Series B keeps rising”]




*Note.* Sample data; replace with your own. Figure adapted from Manning et al. (2008, p. 121) would be cited like this.

Summary

# Two points to carry into the lab

1. [Restate the headline assertion from Part 1]
2. [Restate the headline assertion from Part 2]

## Next steps

**Lab**
[Lab number and task]

**Reading**
[Chapter or pages]

**Due**
[Assessment and date]

**Next lecture**
[Topic]

# Homework

Given Mitchell (1997) definition of machine learning, suppose your email client watches which emails you do or do not mark as spam, and based on that learns how to better filter spam. What is the experience *E* in this setting?

**A** Classifying emails as spam or not spam.

**B** Watching you label as spam or not spam.

**C** The number (or fraction) of emails correctly classified as spam/not spam.

**D** None of the above - this is not a machine learning problem.

Think alone for 1 minute, then compare with a neighbor.

**Next:** Binary search needs sorted data. What does sorting cost?

# Structured Data

- Data for a machine learning task is often recorded in a table.

# References

Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to information retrieval*. Cambridge University Press.

Mitchell, T. M. (1997). *Machine learning*. McGraw-Hill. http://www.worldcat.org/oclc/61321007